1 (a)	$h = 3$	B1
	Area $\approx \frac{3}{2}\{2 + 4.81 + 2\times(3.32 + 4 + 4.46)\}$	M1
	$=$ awrt 45.6	A1
		(3)
(b)(i)	States or uses $\log_2 4x^2 = 2\log_2 2x$	M1
	Hence $\int_2^{14} \frac{\log_2 4x^2}{5}\,dx = \frac{2}{5}\times(a) =$ awrt 18.2	A1 ft
(ii)	Attempts to identify $\log_2 2x$ e.g $\log_2 \frac{2}{x} = \log_2 4 - \log_2 2x = 2 - \log_2 2x$	M1
	Hence $\int_2^{14} \log_2 \frac{2}{x}\,dx = \int_2^{14} (2 - \log_2 2x)\,dx = 24 - 45.6 =$ awrt -21.6	A1ft
		(4)
		(7 marks)

2. (a)	For correct term ${}^6C_4 3^2 (ax)^4$ Sets $15\times 3^2 a^4 = 540$ $a^4 = 4 \Rightarrow a = \pm\sqrt{2}$	M1 A1 dM1 A1 **(4)**
(b)	$\frac{1}{81}\times 3^6 + a^6$ $=17$	M1 A1 A1 **(3)** **(7 marks)**

3. (a)	Attempts $f\left(-\frac{3}{2}\right)=6\left(-\frac{3}{2}\right)^3+17\left(-\frac{3}{2}\right)^2+4\left(-\frac{3}{2}\right)-12$ $=0\Rightarrow(2x+3)$ is a factor *	M1 A1* **(2)**
(b)	$6x^3+17x^2+4x-12=(2x+3)\left(3x^2+4x-4\right)$ $=(2x+3)(3x-2)(x+2)$	M1 A1 dM1 A1 **(4)**
(c)	Solves $\tan\theta=-\frac{3}{2}$ or "–2" or "$\frac{2}{3}$ " $\theta=$ awrt 2.03, 2.16	M1 A1 **(2)** **(8 marks)**

4	$\int\left(2x^2+7\right)\mathrm{d}x=\frac{2}{3}x^3+7x$ or $\int\left(10-2x^2\right)\mathrm{d}x=10x-\frac{2}{3}x^3$	M1 A1
	Achieves/uses a limit of $\sqrt{5}$	B1
	Area $=17\sqrt{5}-\int_0^{\sqrt{5}}\left(2x^2+7\right)\mathrm{d}x$ | Area $=\int_0^{\sqrt{5}}\left(10-2x^2\right)\mathrm{d}x$	M1
	$=17\sqrt{5}-\frac{2}{3}\times5\sqrt{5}-7\sqrt{5}$ | $=10\sqrt{5}-\frac{2}{3}\times5\sqrt{5}$	M1
	$=\frac{20}{3}\sqrt{5}$	A1
		(6) **(6 marks)**

5 (a)	Attempts $30\,000 \times r^3 = 34\,000$ $r^3 = \dfrac{17}{15} \Rightarrow r = 1.0426$ Hence $p = 4.26$	M1 A1 A1ft **[3]**
(b)	Attempts $30\,000 \times (1.0426)^{"N"} = 75\,000$ or with > or < etc throughout. $(1.0426)^{"N"} = \dfrac{5}{2}$ Takes logs $"N" = \dfrac{\log\left(\dfrac{5}{2}\right)}{\log 1.0426}$ (= awrt 21.96) $N = 22$	M1 A1 M1 A1 B1 **[5]** **(8 marks)**

$x^2 + y^2 + 6x - 4y - 14 = 0$

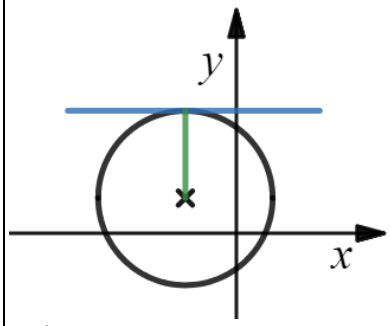
6.(a) Attempts $(x \pm 3)^2 + (y \pm 2)^2 \ldots\ldots (= 0)$ M1

(i) Centre $(-3, 2)$ A1

(ii) Radius $\sqrt{27}$ or $3\sqrt{3}$ A1

(3)

(b)



Attempts $y / k = "2" \pm "3\sqrt{3}"$ M1

Both $k = 2 + 3\sqrt{3}$ and $k = 2 - 3\sqrt{3}$ A1

(2)

Alt:

$y = k \Rightarrow x^2 + 6x + k^2 - 4k - 14 = 0$

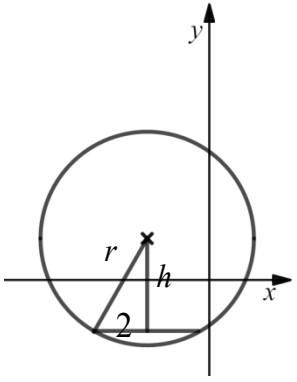
$\Rightarrow b^2 - 4ac = 0 \Rightarrow 6^2 - 4 \times 1 \times (k^2 - 4k - 14) = 0$

$\Rightarrow 4k^2 - 16k - 92 = 0 \Rightarrow k = \ldots$ M1

$k = 2 \pm 3\sqrt{3}$ (*oe*) A1

(2)

(c)



Attempts Pythagoras to find h $\quad h^2 = r^2 - 2^2$ M1

Attempts $p = 2 - h$ dM1

$p = 2 - \sqrt{23}$ (oe) A1

(3)

Alt1: chord touches circle when $x = -1$ (or -5) or need $(x + 3) = \pm 2$ M1

$\Rightarrow 2^2 + (y-2)^2 = 27 \Rightarrow y = \ldots$ or $(-1)^2 + y^2 + 6(-1) - 4y - 14 = 0 \Rightarrow y = \ldots$ dM1

$p = 2 - \sqrt{23}$ A1

(3)

Alt2: Sets $y = p$ and applies difference of roots to be 4

$\Rightarrow x^2 + 6x + p^2 - 4p - 14 = 0 \Rightarrow x = \dfrac{-6 \pm \sqrt{36 - 4(p^2 - 4p - 14)}}{2}$ M1

$\Rightarrow 2\sqrt{23 + 4p - p^2} = 4$ dM1

$\Rightarrow p^2 - 4p - 19 = 0 \Rightarrow p = \ldots$ A1

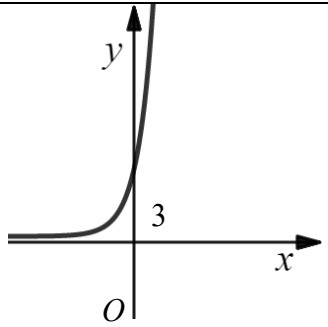
$p = 2 - \sqrt{23}$ **(3)**

(8 marks)

7 (a)	$8\tan\theta = 3\cos\theta$ Uses $\tan\theta = \dfrac{\sin\theta}{\cos\theta} \rightarrow$ $\quad 8\dfrac{\sin\theta}{\cos\theta} = 3\cos\theta$ $8\sin\theta = 3\cos^2\theta$ oe Uses $\cos^2\theta = 1-\sin^2\theta \rightarrow$ $\quad 8\sin\theta = 3\left(1-\sin^2\theta\right)$ $3\sin^2\theta + 8\sin\theta - 3 = 0$ *	M1 M1 A1 *
		(3)
(b)	$(3\sin 2x - 1)(\sin 2x + 3) = 0$ Critical value(s) of $\dfrac{1}{3}, (-3)$ Correct method to find x from their $\sin 2x = \dfrac{1}{3}$ $x =$ awrt $9.74°, 80.26°$	M1 A1 dM1 A1
		(4) **(7 marks)**

8 .(i)	States $(S=)\quad a+\quad (a+d)\quad +.....\{a+(n-2)d\}+\{a+(n-1)d\}$	B1
	$(S=)\{a+(n-1)d\}+\{a+(n-2)d\}+.....(a+d)\quad +a$ and adds	M1
	$2S=n\left(2a+(n-1)d\right)\Rightarrow S=\frac{n}{2}\{2a+(n-1)d\}\quad *$	A1*
		(3)
(ii)	(a) $u_5=22$	B1
	(b) $\sum_{n=1}^{59}u_n=\left(5+10+15+...\right)+\left(-3+3-3+...\right)$	
	$=\frac{59}{2}\{10+58\times5\}+(-3)=8850-3=8847$	M1 B1 A1
		(4)
		(7 marks)

9.(a)



Shape B1

$(0,3)$ B1

(2)

(b)

$$6^{1-x} = 3\times 4^{x}$$

$$(1-x)\log 6 = \log 3 + x\log 4$$ M1 dM1 A1

$$x(\log 4 + \log 6) = \log 6 - \log 3$$

$$\Rightarrow x = \frac{\log\left(\frac{6}{3}\right)}{\log(4\times 6)} \Rightarrow x = \frac{\log 2}{\log 24}$$ ddM1 A1*

(5)

Alt Method: $6^{1-x} = 3\times 4^{x} \Rightarrow \frac{6}{6^{x}} = 3\times 4^{x}$ M1

$$\Rightarrow \frac{6}{3} = 6^{x}4^{x} \Rightarrow 2 = 24^{x}$$ dM1A1

$$\Rightarrow \log_{10} 2 = x\log_{10} 24 \Rightarrow \frac{\log_{10} 2}{\log_{10} 24}$$ ddM1A1*

(5)

(7 marks)

10.(a)	$y = 4x^3 - 9x + \frac{k}{x} \Rightarrow \frac{dy}{dx} = 12x^2 - 9 - \frac{k}{x^2}$	M1 A1
	$x = \frac{1}{2}, \frac{dy}{dx} = 0 \Rightarrow 0 = 12 \times \left(\frac{1}{2}\right)^2 - 9 - 4k \Rightarrow 4k = -6 \Rightarrow k = -\frac{3}{2}$ *	dM1 A1*
		(4)
(b)	Attempts $\frac{d^2y}{dx^2} = 24x - \frac{3}{x^3}$ at $x = \frac{1}{2}$	M1
	$\Rightarrow \frac{d^2y}{dx^2} = -12 < 0$ (Local) Maximum	A1
		(2)
(c)	$12x^2 - 9 + \frac{3}{2x^2} = 0 \Rightarrow 24x^4 - 18x^2 + 3 = 0$	M1
	$\Rightarrow 3\left(4x^2 - 1\right)\left(2x^2 - 1\right) = 0$	A1
	$\Rightarrow x^2 = \frac{1}{2} \Rightarrow x = \frac{\sqrt{2}}{2}$ oe	dM1 A1
		(4)
		(10 marks)